

2.3.1 Limit Laws

We begin this section by stating some intuitively obvious limit laws. We will be able to provide rigorous proofs of these laws at the end of section 2.5.

Theorem: Suppose that c is a constant, n is a positive integer and the limits $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ exist.

Then,

1. $\lim_{x \rightarrow a} c = c$
2. $\lim_{x \rightarrow a} x = a$
3. $\lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$
4. $\lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x)$
5. $\lim_{x \rightarrow a} [cf(x)] = c \lim_{x \rightarrow a} f(x)$
6. $\lim_{x \rightarrow a} [f(x)g(x)] = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x)$
7. $\lim_{x \rightarrow a} \left[\frac{f(x)}{g(x)} \right] = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}$ provided $\lim_{x \rightarrow a} g(x) \neq 0$
8. $\lim_{x \rightarrow a} [f(x)]^n = \left[\lim_{x \rightarrow a} f(x) \right]^n$
9. $\lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \rightarrow a} f(x)}$ provided $\lim_{x \rightarrow a} f(x)$ is positive when n is even

Example 1:

Use the limit laws to calculate the following limit.

a) $\lim_{x \rightarrow 5} [x^2 - 4x + 3]$

b) $\lim_{x \rightarrow -2} \frac{x^3 + 2x^2 - 1}{5 - 3x}$

We see in the examples above that the final limits could be obtained by simply plugging the number x is approaching into the expression. The following theorem makes this idea precise.

Theorem (Direct Substitution Property): If f is a polynomial or a rational function (i.e. the ratio of two polynomial functions) and a is in the domain of f , then $\lim_{x \rightarrow a} f(x) = f(a)$.

Note:

Example 2: Evaluate $\lim_{h \rightarrow 0} \frac{(3+h)^2 - 9}{h}$.

Note: The expression will not always simplify nicely when the direct substitution property cannot be initially applied. The next example illustrates this.

Example 3: Evaluate $\lim_{x \rightarrow 0} \frac{1}{x^2}$

Strategy for evaluating a limit of the form $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$:

1. We first try to plug the value of a in for x . If we get a real number, then that number is our answer. If we get an undefined expression, we proceed to step 2.
2. If the limit of both the numerator and denominator is 0, then it is helpful to algebraically simplify before calculating a limit. Otherwise, we proceed to step 3.
3. If the limit of the denominator is 0 and the limit of numerator is not, then we check to see if/how the limit approaches infinity.

Note: Technically we should have specified in the above strategy that f and g are polynomials. However, we will see later on the strategy may be applied for essentially any functions we will encounter in Calculus one.

Note: This strategy is not perfect! Step 2 is especially not perfect, as sometimes algebraic simplifications will not yield an answer. Later on, we will develop more sophisticated techniques for dealing with the situation where both numerator and denominator approach 0.

Example 4: Evaluate $\lim_{x \rightarrow 7} \frac{\sqrt{x+2}-3}{x-7}$

Example 5: Explain why $\lim_{x \rightarrow \frac{\pi}{2}} \tan(x)$ does not exist.

We now recall an important definition from Precalculus.

Definition: The absolute value of a real number x is: $|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$.

Example 6: Prove that $\lim_{x \rightarrow 0} \frac{|x|}{x}$ does not exist.

Example 7: Calculate the following limit: $\lim_{x \rightarrow 5} \frac{x^2 - 3x - 10}{x^2 - 10x + 25}$

Example 8:

Suppose that f has the following piecewise definition:

$$f(x) = \begin{cases} \frac{1}{x+2} & \text{if } x < -2 \\ x^2 - 5 & \text{if } -2 < x \leq 3 \\ \sqrt{x+13} & \text{if } x > 3 \end{cases}$$

Compute the following limits

a) $\lim_{x \rightarrow 0} f(x)$

b) $\lim_{x \rightarrow 3} f(x)$

c) $\lim_{x \rightarrow -2} f(x)$

Obtaining Limits by Squeezing

General Idea:

Suppose we have three functions where $f(x) \leq g(x) \leq h(x)$ for all values of x near some value a .

If it is the case that $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L$

Then...

The Squeeze Theorem: If $f(x) \leq g(x) \leq h(x)$ when x is near a (except possibly at a) and

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L$$

then,

$$\lim_{x \rightarrow a} g(x) = L$$

Example 9: Show that $\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right) = 0$

Example 10: Find (with justification) $\lim_{x \rightarrow 0^+} (\sqrt{x}) e^{\sin\left(\frac{\pi}{x}\right)}$.